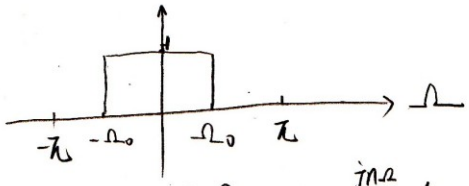
HW #7

(1)

1.



(a) $h(n) = \frac{1}{2\pi}\int_{<2\pi>} H(\Omega) e^{j n\Omega} \cdot d\Omega$

$= \frac{1}{2\pi}\int_{-\Omega_0}^{\Omega_0} 1 \cdot e^{jn\Omega} \cdot d\Omega = \frac{1}{2\pi} \cdot \frac{1}{jn} e^{jn\Omega}\Big|_{-\Omega_0}^{\Omega_0} = \frac{1}{2\pi} \cdot \frac{1}{jn}\left[e^{jn\Omega_0} - e^{-jn\Omega_0}\right]$

$= \frac{1}{2\pi} \cdot \frac{1}{jn} \cdot 2j \sin(n\Omega_0) = \frac{1}{\pi n}\sin(n\Omega_0)$

Recall: $\operatorname{sinc} x = \frac{\sin(\pi x)}{\pi x}$

$$h(n) = \frac{1}{\pi n}\sin(n\Omega_0) = \frac{\sin\left(\pi \cdot \frac{n\Omega_0}{\pi}\right)}{\pi \cdot \frac{n\Omega_0}{\pi}} \cdot \frac{\Omega_0}{\pi} = \frac{\Omega_0}{\pi} \cdot \operatorname{sinc}\left(\frac{n\Omega_0}{\pi}\right) \quad -\infty < n < \infty$$

(b). $\Omega_0 = 0.125\pi$, $F_s = 8KHZ$

the cut off frequency: $f_c = \frac{\omega}{2\pi} = \frac{\Omega_0}{2\pi T_s} = \frac{\Omega_0}{2\pi} \cdot F_s = \frac{0.125\pi}{2\pi} \cdot 8k = 500HZ$

$x_a(t) = 2\cos(150\pi t) + 3\sin(150\pi t + 6)$,

with analog frequency: $f_1 = \frac{150\pi}{2\pi} = 75HZ$ $f_2 = \frac{150\pi}{2\pi} = 75HZ$

so all the input signal can pass through the filter and the output is

$y(n) = x(n) \circledast h(n) \longleftrightarrow X(\Omega) \cdot H(\Omega)$

since $H(\Omega) = 1$ when $|\Omega| < \Omega_0$,

$y(n) = x(n) = x_a(nT_s) = 2\cos(150\pi \cdot n \cdot \frac{1}{8k}) + 3\sin(150\pi \cdot n \cdot \frac{1}{8k} + 6)$

$$= 2\cos\left(\frac{\pi n}{160}\right) + 3\sin\left(\frac{3\pi n}{160} + 6\right)$$

(c) $F_s = 16kHZ$, $\Omega_0 = 0.125\pi$

the cut off frequency: $f_c = \frac{\Omega_0}{2\pi} \cdot F_s = \frac{0.125\pi}{2\pi} \cdot 16K = 1000HZ$

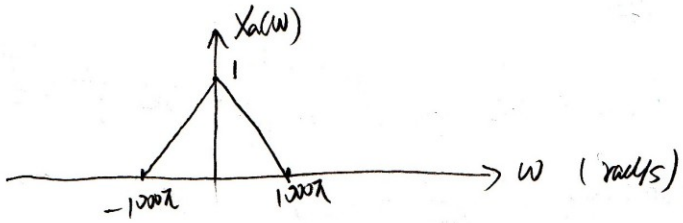
the same procedure as (b).

$y(n) = 2\cos(150\pi n \cdot \frac{1}{16k}) + 3\sin(150\pi \cdot n \cdot \frac{1}{16k} + 6)$

$$= 2\cos\left(\frac{\pi n}{320}\right) + 3\sin\left(\frac{3\pi n}{320} + 6\right)$$

2. $X_a(w) = \begin{cases} 1 - \frac{|w|}{1000\pi} & |w| \le 1000\pi \\ 0 & o.w. \end{cases}$

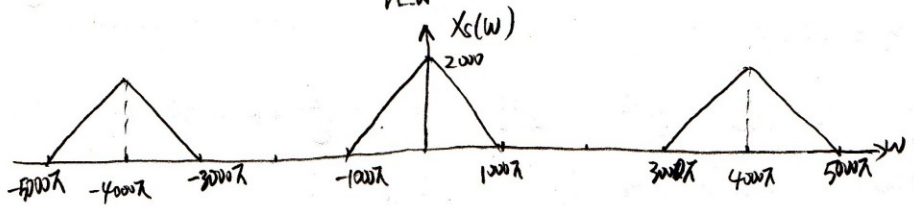
a). $X_a(w)$

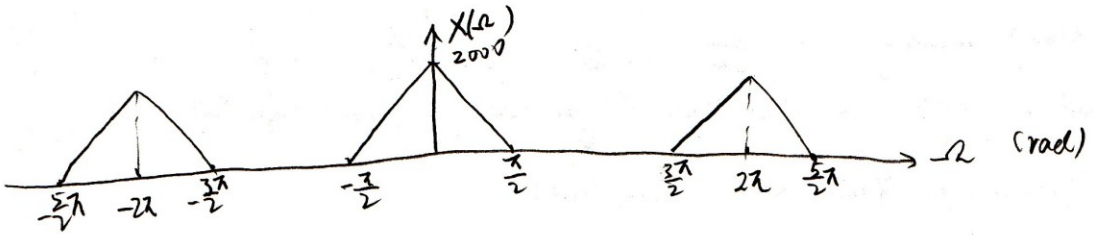


(b). $F_s = 2000\ Hz$, $w_s = \frac{2\pi}{T} = 2\pi \cdot F_s = 4000\pi$

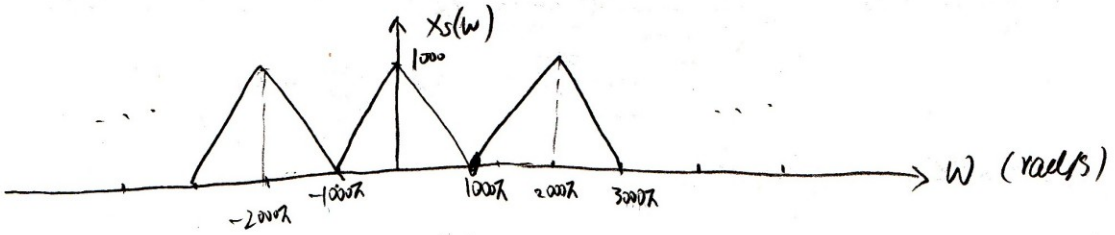
$X_s(w) = \frac{1}{T}\sum_{n=-\infty}^{\infty} X_a(w - nw_s)$ $\quad X(\Omega) = \frac{1}{T}\sum_{n=-\infty}^{\infty} X_a(\frac{\Omega}{T} - nw_s)$

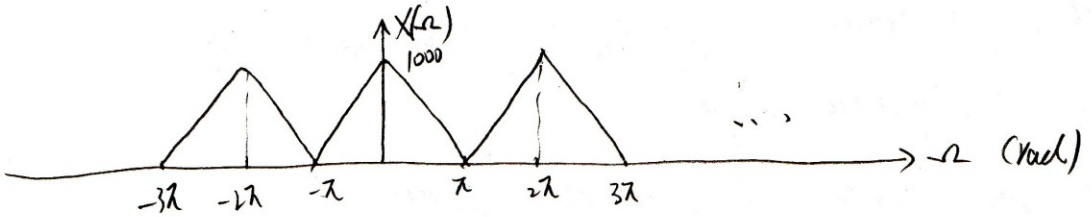
$= 2000 \cdot \sum_{n=-\infty}^{\infty} X_a(w - 4000\pi)$ $\quad = X_s(\frac{\Omega}{T})$ $\quad w = \frac{\Omega}{T}$





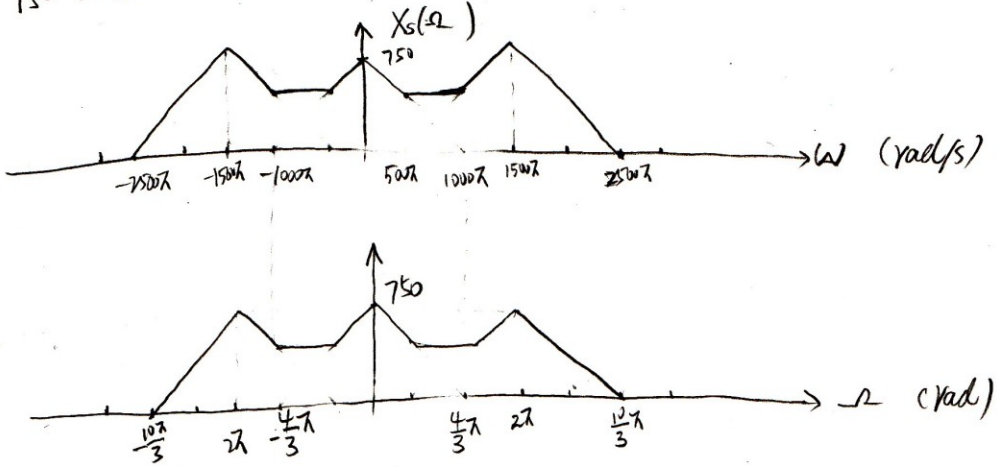
(c) $F_s = 1000Hz$. $w_s = 2\pi \cdot F_s = 2000\pi$





(3)

(a) $F_s = 750Hz$ $\quad \omega_s = 2\pi F_s = 2\pi \cdot 750 = 1500\pi$



3. (a) $H(\Omega) = \dfrac{\frac{1}{2} + \frac{1}{12}e^{-j\Omega}}{1 + \frac{5}{8}e^{-j\Omega} + \frac{1}{8}e^{-j2\Omega}}$ $\quad$ let $z = e^{j\Omega}$

$$H(z) = \frac{\frac{1}{2} + \frac{1}{12}z^{-1}}{1 + \frac{5}{8}z^{-1} + \frac{1}{8}z^{-2}} = \frac{6z^2 + z}{12z^2 + 10z + 2} = \frac{z(6z+1)}{2(6z^2+5z+1)} = \frac{\frac{1}{2}z(z+\frac{1}{6})}{(z+\frac{1}{2})(z+\frac{1}{3})}$$

$$\frac{H(z)}{z} = \frac{A_0}{(z+\frac{1}{2})} + \frac{A_1}{(z+\frac{1}{3})}$$

$$A_0 = (z+\tfrac{1}{2})\frac{H(z)}{z}\Big|_{z=-\frac{1}{2}} = \frac{\frac{1}{2}(z+\frac{1}{6})}{z+\frac{1}{3}}\Big|_{z=-\frac{1}{2}} = \frac{\frac{1}{2}(-\frac{1}{2}+\frac{1}{6})}{-\frac{1}{2}+\frac{1}{3}} = 1$$

$$A_1 = (z+\tfrac{1}{3})\frac{H(z)}{z}\Big|_{z=-\frac{1}{3}} = \frac{\frac{1}{2}(z+\frac{1}{6})}{z+\frac{1}{2}}\Big|_{z=-\frac{1}{3}} = \frac{\frac{1}{2}(-\frac{1}{3}+\frac{1}{6})}{-\frac{1}{3}+\frac{1}{2}} = -\frac{1}{2}$$

$$H(z) = \frac{z}{z+\frac{1}{2}} - \frac{1}{2}\cdot\frac{z}{z+\frac{1}{3}}$$

$$\Rightarrow \boxed{h(n) = \left[(-\tfrac{1}{2})^n - \tfrac{1}{2}\cdot(-\tfrac{1}{3})^n\right]u(n)}$$

(b) $$\frac{Y(z)}{X(z)} = H(z) = \frac{\frac{1}{2} + \frac{1}{12}z^{-1}}{1 + \frac{5}{8}z^{-1} + \frac{1}{8}z^{-2}}$$

$$\Rightarrow \boxed{y(n) + \tfrac{5}{8}y(n-1) + \tfrac{1}{8}y(n-2) = \tfrac{1}{2}x(n) + \tfrac{1}{12}x(n-1)}$$